

Amirhossein Nafissi

226-600-2648 | nafissi.amir@gmail.com | amirnafissi.com | linkedin.com/in/amir-nafissi | github.com/Amir-Nafissi

EDUCATION

University of Waterloo

Bachelor of Software Engineering (BSE)

Waterloo, ON

Sep. 2025 – Jan. 2030

University of Guelph

Bachelor of Computing, Software Engineering (Transferred); GPA: 4.0/4.0

Guelph, ON

Sep. 2024 – April 2025

TECHNICAL SKILLS

Languages: Python, C, C++, Java

Frameworks & Libraries: Pandas, Matplotlib, NumPy

Databases & Infrastructure: MongoDB, Vercel

Developer Tools: Git, Bash, VS Code, Jira, Agile/Scrum

AI-Assisted Development: Claude Code (Hooks, Workflows, Orchestration)

EXPERIENCE

TechOps Developer

May 2026 – August 2026

Instant Risk Coverage (IRC)

Toronto, ON

- Architected and developed a Claude Code-powered PR review tool using shell scripts and Mermaid.js to visualize system architecture and align business and engineering rules, accelerating pull request review cycles.
- Developed a data pipeline using Mage, SQL, and Python for storing and processing customer orders, facilitating IRC's next-generation product development and enabling streamlined reporting.
- Engineered USD payment acceptance for IRC's existing insurance portals, expanding the platform's market reach.
- Managed technology operations including client onboarding, cancellation processing, and iterative improvements to IRC's insurance portal products, ensuring reliable service delivery.

Founding Member & Software Developer

Nov. 2025 – Present

WatQ – Waterloo Quantum Design Team

Waterloo, ON

- Designed and built the team's public-facing website using Next.js, React Three Fiber, and Framer Motion featuring an interactive 3D Bloch sphere.
- Architected the team's Notion workspace as the single source of truth, reducing information fragmentation and accelerating onboarding.
- Recruited and secured the team's academic advisor, establishing faculty mentorship and institutional credibility.

Machine Learning Intern

Jul. 2023 – Oct. 2023

University of Waterloo

Waterloo, ON

- Worked with a research group to collect a sensitive security dataset for training a robust machine learning model.
- Gained training and hands-on experience related to creating simple machine-learning models with various Python libraries to be used in personal projects.

PROJECTS

CueMate | Kotlin, Jetpack Compose, MediaPipe, CameraX

- Developed a privacy-first Android assistive application that leverages on-device computer vision to interpret facial expressions and hand gestures for visually impaired users.
- Integrated Google's MediaPipe framework to run real-time facial landmark analysis and gesture recognition locally, providing multimodal feedback via Text-to-Speech and directional haptics.

Visualize Quantum | Next.js, TypeScript, Python, NumPy/SciPy, Qiskit, Tailwind CSS

- Engineered a modular quantum algorithm playground with a plugin architecture (AlgorithmModule interface + central registry), enabling new algorithms to drop in as self-contained modules without modifying the core framework.
- Implemented a Graph Coloring solver using a hand-written NumPy statevector QAOA (COBYLA optimizer, layerwise INTERP warm-starting) for exact local simulation, with an IBM Quantum hardware path via Qiskit for sampling optimized circuits on real backends.
- Built interactive visualizations—force-directed graph canvas, energy-convergence charts, and a conflict-distribution chart showing probability mass by conflict count—deployed as a serverless Python function on Vercel with a warm-worker dev server.